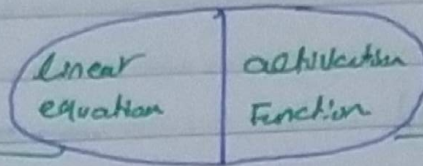


Deep learning

→ Newton :-



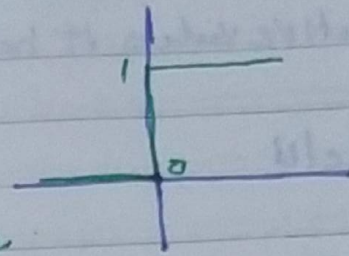
$$y = \sum_{i=0}^n w_i x_i + b$$

segmoied - Tanh - Relu

→ Activation Functions :-

- Step Function 0 / 1

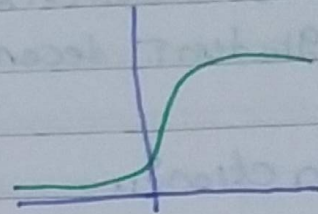
• The Output can have more than one (1)
so it causes Problems in classification



- segmoied Function 0 → 1

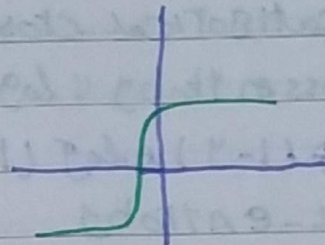
$$Z = \frac{1}{1 + e^{-y}}$$

solved the step Problem



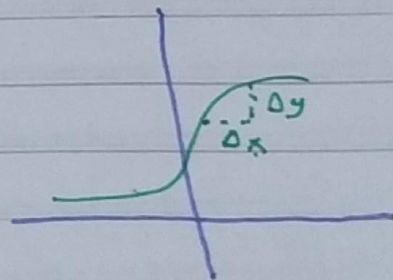
- Tanh Function -1 → 1

$$Z = \frac{e^y - e^{-y}}{e^y + e^{-y}}$$



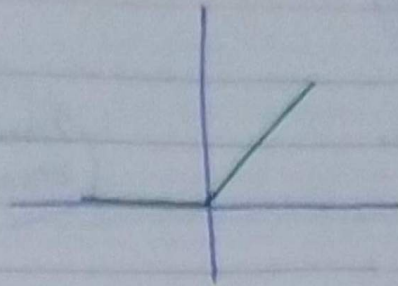
* Mostly, we use segmoied in the output layer and tanh in all other layers, as it's calculated mean is zero so it centers out data

Tanh & segmoied has Problem that for high values $\frac{\Delta y}{\Delta x} = \text{zero}$ so learning Rate is zero



→ ReLU Function

$$\text{ReLU}(z) = \max(0, x)$$



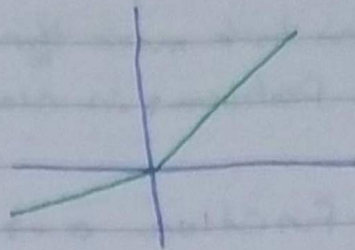
for hidden layers ReLU is mostly popular used function if we not sure what function to use just use ReLU as default choice

|| for negative values it has the same learning rate problem ||

→ Leaky ReLU

$$\text{leaky ReLU}(z) = \max(-1x, x)$$

solves the gradient decent Problem



→ Loss Functions:

→ sparse categorical cross entropy

→ binary cross entropy || log loss || used for logistic regression

$$-\log(\hat{y}) - (1-y) \cdot \log(1-\hat{y})$$

→ categorical entropy

→ mean absolute error $|\hat{y} - y|$

→ mean square error $(\hat{y} - y)^2$ allows gradient descent to converge in better way ||